Problem 1. (5+5+4+4+2 = 20 points) Let $A = (2, -5, 6)$, $B = (3, -5, 7)$, and $C = (3, -6, 8)$.
(a) Compute $\vec{AB} \times \vec{AC}$ and $\vec{AB} \cdot \vec{AC}$.

(b) Find the angle between the vectors $\vec{AC}$ and $\vec{AB}$.

(c) Find a parametric equation for the line which contains the point B and is parallel to the vector $\vec{AC}$

(d) Find the equation for the plane containing the points A, B, and C.

(e) True or False: The line defined in part (c) must lie in the plane defined in part (d). Justify your answer.

Problem 2. (5 + 5 = 10 points) Consider the curve $\mathcal{C}$ parametrized by

$$\mathbf{r}(\theta) = \langle \sin^3(\theta), \cos^3(\theta), \sin^2(\theta) \rangle$$

for $0 \leq \theta \leq \pi/2$.

(a) Provide a parametric equation for the line tangent to $\mathcal{C}$ at the point $\left(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, \frac{1}{2}\right)$.

(b) Compute the length of C.

Problem 3. $(5+5+5+5=20$ points$)$ Consider the function $f(x, y)=y^2-x^2-yx^3$.
(a) Compute $\nabla f(1,-1)$.

(b) Find an equation for the tangent plane to the graph of $z = f(x, y)$ at the point $(1, -1, 1)$.

(c) Compute the directional derivative of f at the point $(1, -1)$ in the direction of $\langle 3, 1 \rangle$.

(d) True or false: The level curve defined by $f(x,y) = 1$ has, at the point $(1,-1)$, tangent line parallel to $\langle 3, 1\rangle$. Justify your answer.

Problem 4. (10+5+5 =20 points) Throughout this problem $f(x,y)=x^2+y^2-xy$.
(a) Let S^1 denote the unit circle in the plane, that is, $S^1=\{(x,y)\,|\,x^2+y^2=1\}$. Use the method of Lagrange multipliers to find the maximum and minimum values obtained by $f(x,y)$ on S^1.

(b) Find the critical point of $f(x, y)$.

(c) Let D denote the unit disk; that is, $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. Combine parts (a) and (c) to determine the maximum and minimum values for $f(x, y)$ on D.

Problem 5. (5+7+8=20 points) Throughout this problem $f(x, y) = x^3 + y^2 - 6xy + 6x + 3y$.
(a) Compute ∇f at an arbitrary point (x, y).

(b) Find all critical points of $f(x, y)$.

(c) Use the second derivatives test to classify the critical points found in part (b).

Problem 6. (2x5=10 points) Let $f(x,y) = e^{x-y}$.
(a) Suppose x and y are functions of t with $x(3) = y(3) = 1/2$ and $x'(3) = 2$ while $y'(3) = 4$. Compute $\frac{df}{dt}(3)$.

(b) Suppose x and y are functions of s and t with $x(3,2) = 7$, $y(3,2) = -5$, $x_s(3,2) = 8$, $x_t(3,2) = 1$, $y_s(3,2) = 5$ and $y_t(3,2) = -1$. Compute the partial of f with respect to t at the point $(3,2)$ (that is, the point where $s = 3$ and $t = 2$).